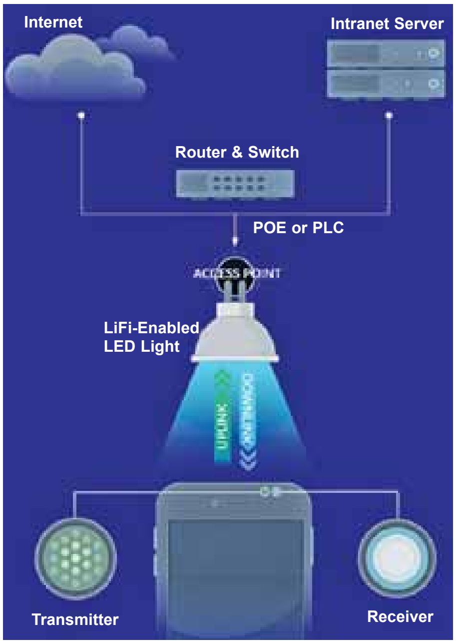
## Li-Fi—the world's fastest Wi-Fi

Li-Fi is a wireless internet communication system that uses visible light for transmission at very high speeds. Combining the world of electronics and photonics, this technology captures data in modulated light frequencies of an LED light source and transmits them. The photosensitive detector converts the light frequency signal back to an electronic data stream, allowing faster bi-directional wireless communication.

Like decoding Morse code, this technology has a much faster rate of decoding information, which is up to millions of times per second. In terms of transmission speed, Li-Fi can go over 100Gbps, which is nearly 14 times faster than even WiGig.




*The basics of a Li-Fi system (Credit: https://Li-Fi.co/what-is-Li-Fi)*


*Fig. 4: Geographical location*

*Fig. 5: Optical communication unit for Li-Fi*

Since the Ice Stupa is in a conical shape, it can store a great amount of water, offering a very less surface area for evaporation. As the Ice Stupa slowly melts, the water is stored in a tank and is used for farming using the drip irrigation method.

Using Li-Fi technology, Sonam Wangchuk conducted a live Youtube session on April 1, 2022 and displayed the speed of the internet.

Sonam adds, "It is simple with no high-technology power. It has become a movement in villages with competitions being held nowadays for the largest and biggest Ice Stupas. This innovation is expanding its borders and spilling outside India into Nepal, Pakistan, Chile, Switzerland, Czech Republic, Europe, and so on—both for aesthetics and winters as well."

### Mountain-to-mountain connectivity

The top of a mountain is usually isolated and cut-off with no internet connectivity found. But mountains are the mightiest and biggest structures that can be used to erect towers. Combining this with an interesting technology called Li-Fi, the mountain-to-mountain connectivity can be achieved.

In terms of geography, the signal from the Jio tower cannot be received nor seen at the SECMOL school. However, it can be seen and received on the SECMOL mountain (Li-Fi Tower1 hub), which is at an altitude of 800m. After the signal is captured at the Li-Fi Tower1 hub as a line-of-sight (LOS) propagation from the Jio tower, it transmits the signal to the optical communication unit (Fig. 6) via light modulation (laser beams). This unit is placed on the roof of the SECMOL school and the signal from the Li-Fi Tower1 hub is decoded and converted back to an electrical signal. Hence the signal from the tower can reach remote areas (indirectly), as seen in Fig. 6.

Li-Fi eliminates the cost for digging of optical fibre cables and building of 10-story tall towers. However, powering the towers can be a challenge considering the cold atmosphere on the mountain top. Major telecom providers say that the